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Box Office Analysis for New Movie Studio Presentation

Introduction

Title: Box Office Analysis
for New Movie Studio

Subtitle: Data-Driven
Insights for better Movie
Production.

BY:GROUP 1



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Project overview

The background information that we have is:

The company plans to enter the movie industry by launching a new film studio.

Major competitors are successfully investing in original video content.

There is limited internal knowledge about what drives box office success.

The goal is to analyze current trends and deliver actionable insights to guide film production decisions.

Problem Statement

- **Factors Influencing Box Office Performance:**
- **Unclear Investment Strategy** – Movie studios lack data-driven insights on which factors (e.g., genre, budget, release timing) most impact box office success, leading to risky investment decisions.
- **High Financial Risk** – Without understanding key success drivers, the new studios risks allocating budgets inefficiently, resulting in underperforming films and lost revenue.
- **Missed Market Opportunities** – Failure to optimize release timing or genre trends may cause films to underperform despite strong content, missing peak audience demand.

Project Objectives.

- Determine top-performing genres.
- Find optimal budget ranges.
- Identify the best release months.
- Deliver actionable business recommendations.



Key Findings from the data

- Drama, Documentary, and Comedy are the most frequently produced genres, indicating high market presence and audience demand.
- Action, Adventure, and Fantasy films dominate total gross revenue, highlighting their commercial success.
- Hybrid genres like Comedy–Documentary–Fantasy receive the highest average ratings (above 9), showing strong viewer appreciation and critical acclaim.
- Top-grossing films are primarily from franchises (e.g., Marvel, Star Wars), suggesting sequels and universe-building boost earnings.
- Movies released during peak seasons (June–July, November–December) tend to perform better at the box office.

Data Understanding



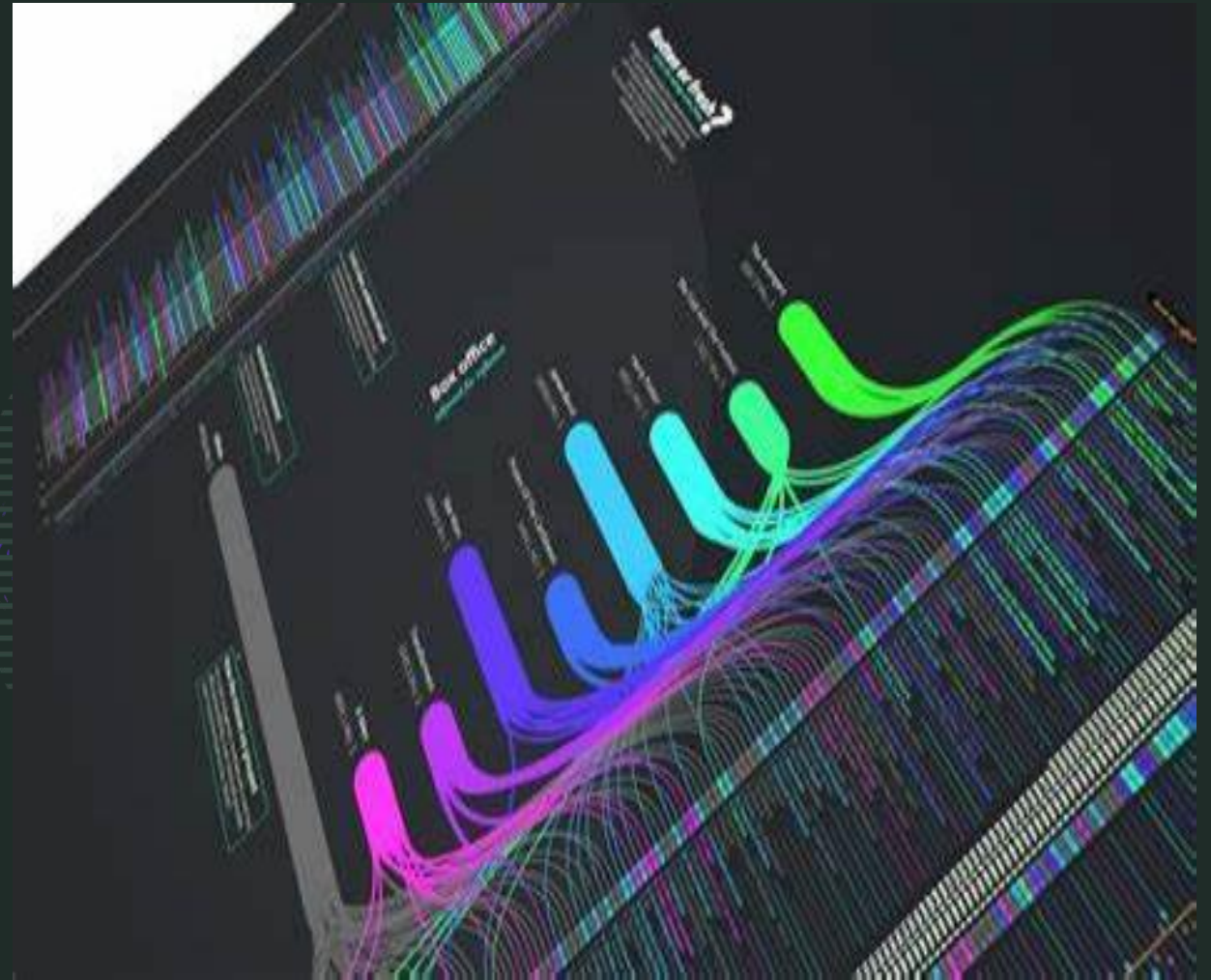
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- Data was collected from trusted movie sources (Box Office Mojo, IMDb, The Numbers).
- Files used included `bom.movie_gross.csv` and `im.db`.
- Key features analyzed: genres, budgets, ratings, revenue, and release dates.

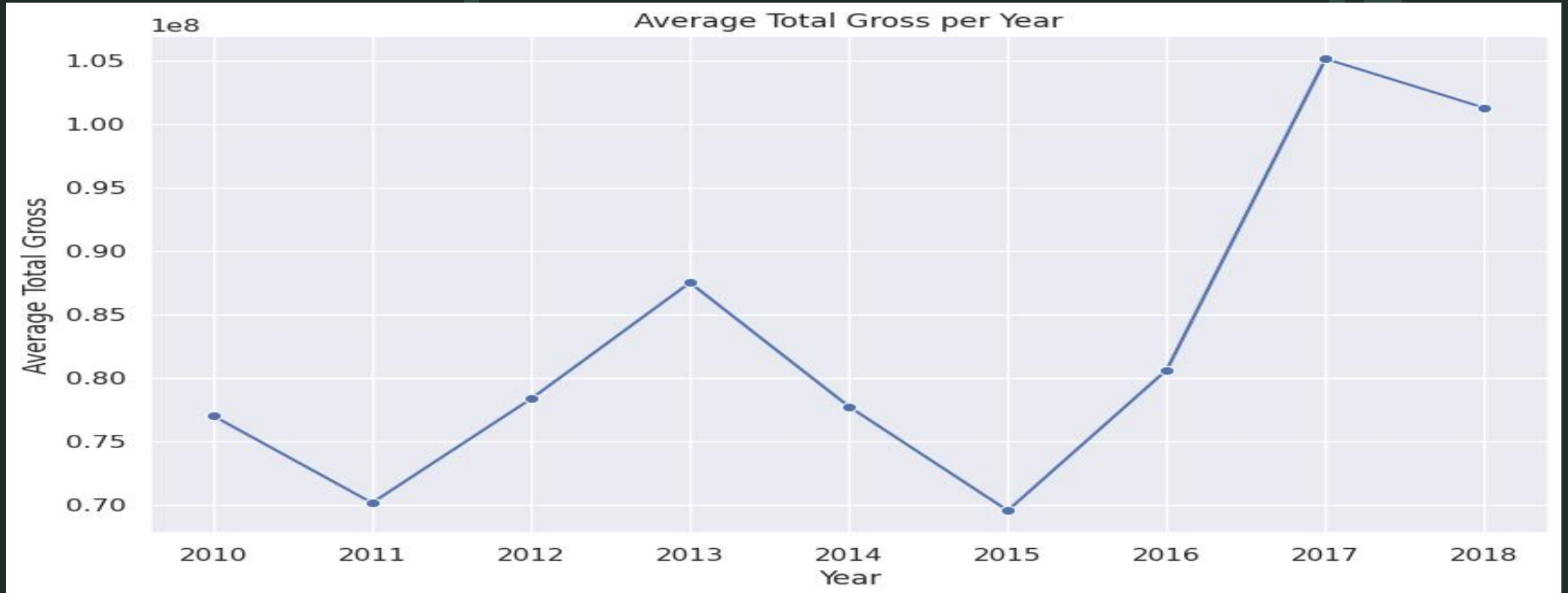
Exploratory Data Analysis (EDA)

visualization

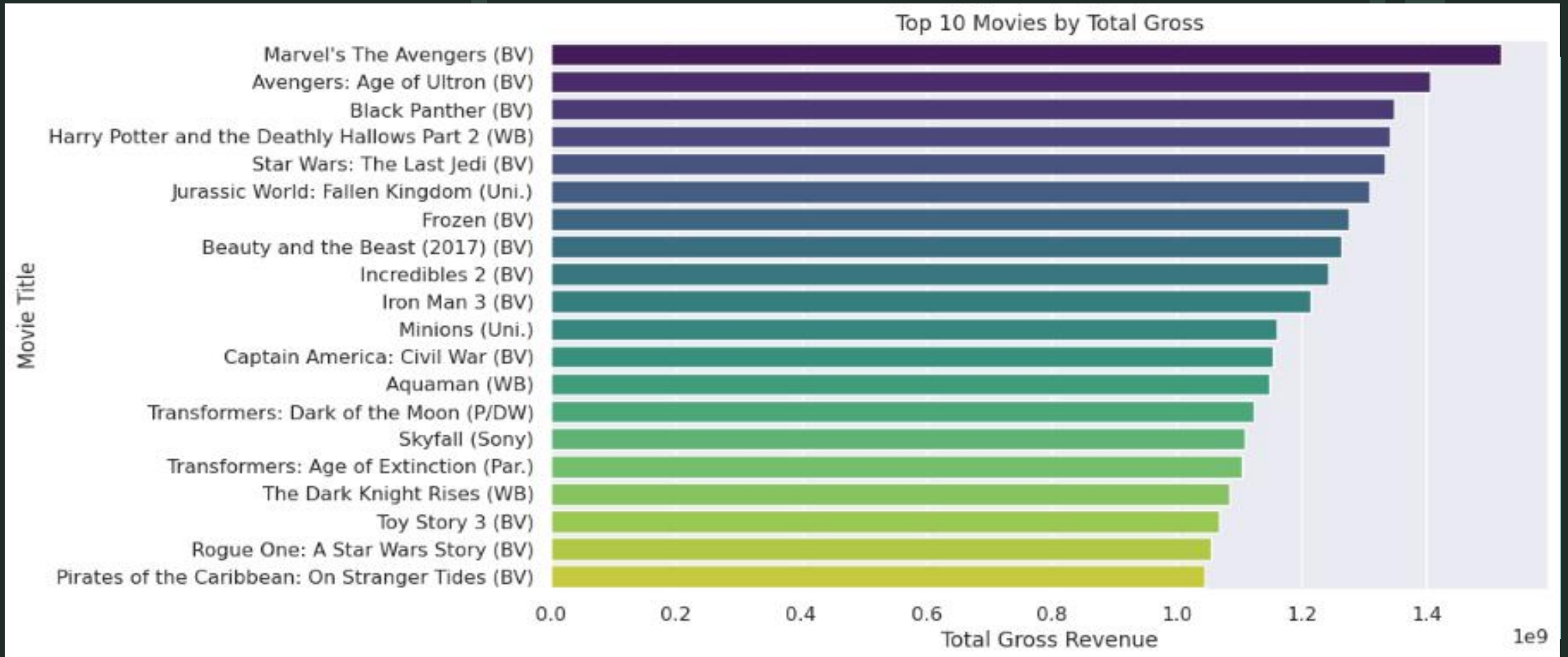
Here we will see some
visualization examples.



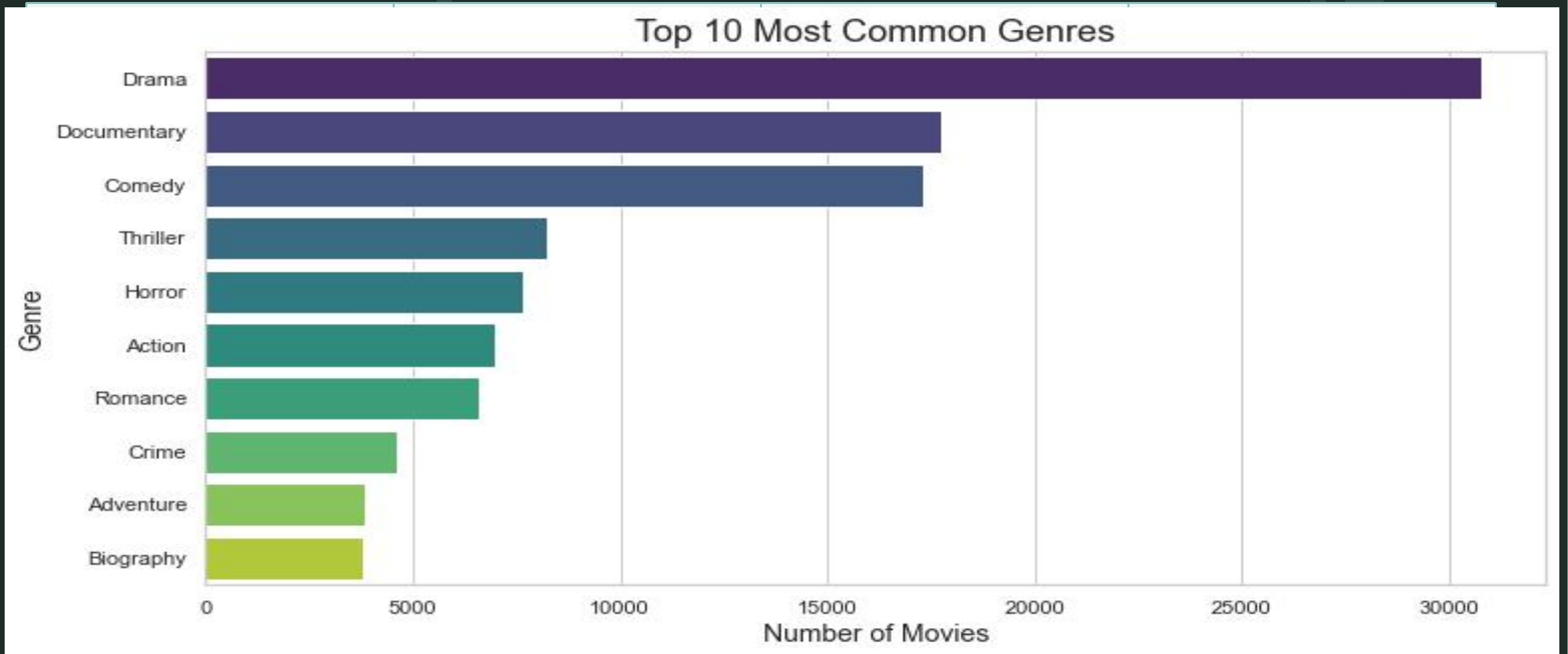
Average Gross per year.



Top ten Movies by Total Gross.



Top ten most common Genres



Data Preparation and Tools

- Libraries like Pandas, NumPy, and SQLite3 were imported for data handling and analysis.
- Files were unzipped and loaded into Pandas data frames.
- Data was cleaned by removing duplicates and handling missing entries.

Cleaned the data by:

- Removing duplicates.
- Handling missing values.
- Correcting inconsistent genres and dates.
- Prepared clean, analysis-ready datasets for further exploration.

Conclusion

- Mid-budget films tend to be more profitable than low-budget or high-budget films, offering a better return on investment.
- Mainstream genres like Action, Comedy, and Adventure consistently perform better at the box office.
- Releasing films during peak seasons (e.g., summer or holidays) significantly boosts earnings and audience reach.
- A strategic combination of genre, budget, and release timing increases the likelihood of a movie's commercial success.

Recommendations

- Create genre-budget criteria to help make decisions more quickly and prevent overspending or underspending on genres that don't work together.
- Use past data trends to predict future performance and direct investment in future motion picture productions.
- Before completing production, use surveys or pilots to gauge audience preferences, particularly for stories that merge genres or are unusual.
- In order to maximize pre-release hype and match promotions with audience availability, marketing activities should be optimized around peak seasons.

Next Steps

- WE PLAN TO:
- To learn more about the combinations of genre, money, and timing that produce top-tier outcomes.
- To measure the degree to which release date, genre, and budget affect box office performance in your dataset.
- Examine trends related to a given genre over time to determine which genres are becoming more or less popular.
- Using a grading system, rank forthcoming movies according to their genre, budget, and release date in order to forecast their likelihood of success.



Thank you

GROUP 1
DSFT12- HYBRID CLASS